



IP Internet Protocol (Location) } device
 MAC Media access control (Identification)

MAC

- Media access control
- It is a physical address ✓
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- $12 \times 4 = 48$ bit length

NIC - MAC / 48 bit

1C-C1-DE-33-8D-15

(Binary)

16 bit

24 bit

Vendor id

Organization

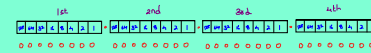
IP

- Internet protocol
- It is a logical address
- It is in dotted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- $8 \times 4 = 32$ bit length
- IP = network + host

NIC - IP

IP Structure

32 bit



IPv4

Sliding IP



Binary IP

Class A B C D E

Types of Communication

unicast one - one
 multicast one - Group
 Broadcast one - All

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Broadcast R&D

Invalid IP

$80.0.0.10 \rightarrow A \rightarrow N.H.H.H$
 $172.80.254.128 \rightarrow B \rightarrow N.N.H.H$
 $192.255.12.82 \rightarrow C \rightarrow N.N.N.H$

School

4th std

2 bit

2 1

0 0

0 1

1 0

1 1

}

3 bit

4 2 1

0 0 0

0 0 1

0 1 0

0 1 1

1 0 0

1 0 1

1 1 0

1 1 1

5 bit

16 8 4 2 1

Pen ~ stop Binary Combinator

Pending

50.0.0.10

N. H. H. H

Network = Street No

Host = Door No

50.50.50.50 IP

172.182.210.32
N. N. H. H

192.192.192.192
N. N. N. H

How many network bit?

24

How many network octet?

3

How many host octet?

1

How many host bit?

8

Fresher

Job

90%

(1 hour)

(Note)

1→